Causal identification in macro III

Version 1 (April 30, 2026)

ECON 31730, Spring 2026
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Proxies

$$y_t = \Theta(L)\varepsilon_t, \quad \varepsilon_t \sim WN(0, I_{n_\varepsilon})$$

- As mentioned in the previous slides, a common empirical strategy nowadays is to construct shocks using a "narrative approach". Then treat shock $\varepsilon_{1,t}$ as observed.
- But probably unreasonable to assume that this procedure picks up *all* the movements in this shock without measurement error.
- More reasonable to assume that our measure z_t is a **proxy** for the true shock:
$$z_t = \alpha\varepsilon_{1,t} + v_t, \tag{†}$$
where the measurement error v_t is classical: white noise and uncorr. with ε_t at all leads/lags.
- Could easily allow z_t to also depend on lags of y_t and z_t; then just interpret (†) as the residual after projecting on those controls.

Proxy-SVMA model

$$y_t = \Theta(L)\varepsilon_t,$$
$$z_t = \alpha\varepsilon_{1,t} + v_t,$$
$$(\varepsilon_t', v_t)' \sim WN(0, \text{diag}(1, \dots, 1, \sigma_v^2))$$

- A proxy variable z_t is also called an **external instrumental variable (IV)** for reasons that will soon become clear.
- Model invokes **relevance and exclusion restrictions** on the IV:

$$E(z_t\varepsilon_{1,t}) = \alpha \neq 0, \quad E(z_t\varepsilon_{j,t}) = 0,\ j \geq 2,$$

$$E(z_t\varepsilon_\tau) = 0_{n_\varepsilon \times 1},\ \tau \neq t.$$

- Normalization: $\alpha > 0$.
- We do *not* assume (partial) invertibility or $n = n_\varepsilon$!

Proxy-SVMA: identification of relative impulse responses

$$y_{i,t} = \sum_{j=1}^{n_\varepsilon} \sum_{\ell=0}^{\infty} \Theta_{i,j,\ell}\varepsilon_{j,t-\ell}, \quad z_t = \alpha\varepsilon_{1,t} + v_t$$

- Without the IV z_t we know we can't identify impulse responses.
- The IV allows us to identify *relative* IRs to the first shock:

$$\text{Cov}(y_{i,t+\ell}, z_t) = \Theta_{i,1,\ell}\,\text{Cov}(\varepsilon_{1,t}, \alpha\varepsilon_{1,t} + v_t) = \alpha\Theta_{i,1,\ell}$$

$$\implies \frac{\text{Cov}(y_{i,t+\ell}, z_t)}{\text{Cov}(y_{1,t}, z_t)} = \frac{\Theta_{i,1,\ell}}{\Theta_{1,1,0}}.$$

- Left-hand side: estimand of a two-stage least squares (2SLS) regression of $y_{i,t+\ell}$ on $y_{1,t}$, using z_t as an IV.
- Recall: No need for invertibility or $n = n_\varepsilon$!
- *Absolute* IRs $\Theta_{i,1,\ell}$ identified up to scalar multiple α (more soon).

LP-IV

- 2SLS version of local projection (**LP-IV**, Stock & Watson, 2018):

$$y_{i,t+\ell} = \hat{\mu}_\ell + \hat{\beta}_\ell y_{1,t} + \hat{e}_{t+\ell|t}$$

  using z_t as an IV for $y_{1,t}$. Repeat for all horizons $\ell \geq 0$.

- $(\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots)$ consistent estimator of $(\frac{\Theta_{i,1,0}}{\Theta_{1,1,0}}, \frac{\Theta_{i,1,1}}{\Theta_{1,1,0}}, \frac{\Theta_{i,1,2}}{\Theta_{1,1,0}}, \dots)$.
- If the IV is weak ($\alpha \approx 0$ or $\Theta_{1,1,0} \approx 0$), usual asymptotic distribution theory can be misleading.
- As usual, we could include exogenous controls (uncorrelated with $\varepsilon_{1,t}$, such as lags of y_t or z_t) for efficiency reasons.
- If we need to do joint inference on multiple IR horizons, could set this up as a linear GMM system.

Application: high-freq. identification of monetary shocks

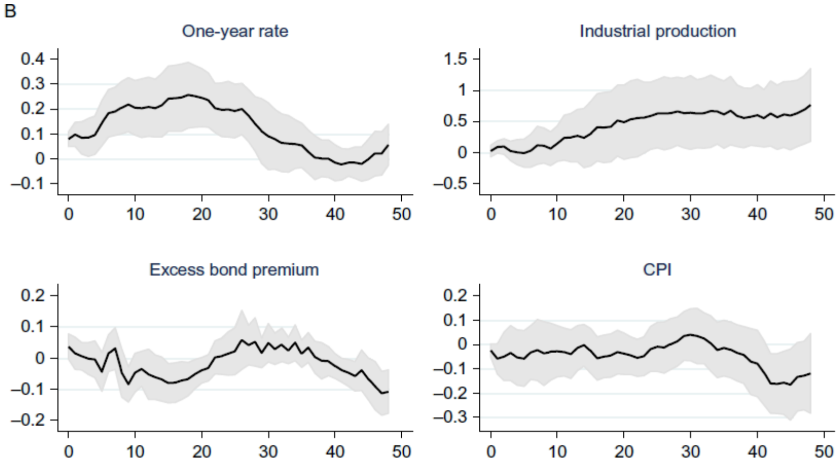

Fig. 3 Gertler–Karadi's monetary shock. (A) Gertler–Karadi's monetary proxy SVAR, VAR from 1979m7 to 2012m6, instrument from 1991m1 to 2012m6. (B) Gertler–Karadi monetary shock, Jordà 1990m1–2012m6. Light gray bands are 90% confidence bands.

Source: Ramey (2016)

VAR implementation of LP-IV

$$y_{i,t+\ell} = \hat{\mu}_\ell + \hat{\beta}_\ell y_{1,t} + \text{controls} + \hat{\mathrm{e}}_{t+\ell|t} \quad \text{with } z_t \text{ as IV}$$

- Consider a version of LP-IV 2SLS where we control for all lags of $w_t = (z_t, y_t')'$ in the regressions.
- As usual for 2SLS with a single endogenous regressor and single IV, the 2SLS estimand is then given by

$$\beta_\ell^{\text{2SLS}} \equiv \frac{\beta_\ell^{\text{RF}}}{\beta^{\text{1st}}},$$

  where we define the "reduced-form" and "first-stage" projections

$$y_{i,t+\ell} = \mu_\ell^{\text{RF}} + \beta_\ell^{\text{RF}} z_t + \textstyle\sum_{h=1}^{\infty} \delta_{\ell,h}^{\text{RF}\prime} w_{t-h} + e_{t+\ell|t}^{\text{RF}},\ \ell = 0, 1, 2, \ldots$$

$$y_{1,t} = \mu^{\text{1st}} + \beta^{\text{1st}} z_t + \textstyle\sum_{h=1}^{\infty} \delta_h^{\text{1st}\prime} w_{t-h} + e_{t|t}^{\text{1st}}.$$

VAR implementation of LP-IV (cont.)

$$\beta_\ell^{\text{2SLS}} = \frac{\beta_\ell^{\text{RF}}}{\beta^{\text{1st}}},$$

$$y_{i,t+\ell} = \mu_\ell^{\text{RF}} + \beta_\ell^{\text{RF}} z_t + \textstyle\sum_{h=1}^{\infty} \delta_{\ell,h}^{\text{RF}\prime} w_{t-h} + e_{t+\ell|t}^{\text{RF}},\ \ell = 0, 1, 2, \dots$$

- The 2SLS IRF estimand is *proportional* to $(\beta_0^{\text{RF}}, \beta_1^{\text{RF}}, \beta_2^{\text{RF}}, \dots)$.
- Hence, if all we want are *relative* IRFs, then we can just run the reduced-form projections and normalize our IRFs appropriately.
- But we have seen already that $(\beta_0^{\text{RF}}, \beta_1^{\text{RF}}, \beta_2^{\text{RF}}, \dots)$ is (proportional to) the IRF in a recursive SVAR(∞) in the data w_t where we order the IV z_t first (Plagborg-Moller & Wolf, 2021).
- Alternative LP-IV estimation strategy: Order the IV first in a VAR and compute (relative) impulse responses to the first shock as usual.

VAR implementation of LP-IV (cont.)

- By the previous LP-IV arguments this VAR strategy works even if ε_1 is non-invertible! Why?
- The shock ε_1 is still possibly non-invertible with respect to the information set $\{y_\tau, z_\tau\}_{-\infty<\tau\leq t}$ in the VAR.
- But the only remaining source of non-invertibility is the measurement error v_t in the IV: $z_t = \alpha\varepsilon_{1,t} + v_t$.
- This measurement error causes attenuation bias in the impulse responses, but the bias turns out to be the *same* (percentage-wise) for all horizons and variables.
- Thus, *relative* impulse responses are correctly estimated. Can do inference on these using standard VAR methods.
- However: We cannot trust the FVD/HD from this VAR without further assumptions. We don't know how much noise there is in z_t.

Proxy-SVMA: identification of VD/FVD/HD

$$y_t = \Theta(L)\varepsilon_t,$$
$$z_t = \alpha\varepsilon_{1,t} + v_t,$$
$$(\varepsilon_t', v_t)' \sim WN(0, \text{diag}(1, \ldots, 1, \sigma_v^2))$$

- Relative impulse responses are very useful. But we would also like to identify VD/FVD/HD to gauge the *importance* of $\varepsilon_{1,t}$.
- Unfortunately, turns out that these are not point-identified without further assumptions.
- Intuitively, we do not know the IV signal-to-noise ratio α^2/σ_v^2. The observed correlation between y_i and z could be consistent with:
  1. Large effect of ε_1 on y_i but small α. OR...
  2. Small effect of ε_1 on y_i but big α.
- Slide appendix: We can derive *bounds* on objects of interest.

Recoverability

- For now, we will impose an extra assumption that turns out to give us point identification of our objects of interest.

- We say that ε_j is **recoverable** if

$$\varepsilon_{j,t} \in \overline{sp}(y_\tau, -\infty < \tau < \infty),$$

  i.e., if $\varepsilon_{j,t}$ can be obtained as a (linear) function of all past, present, and *future* values of the data $\{y_t\}$.

- Clearly weaker than assuming (partial) invertibility.

- If $n = n_\varepsilon$, then we automatically have recoverability of *all* shocks, under the additional condition that $s_y(\omega)$ is non-singular everywhere. This is because there then exists an abs. summ. *two-sided* inverse $\Theta(L)^{-1}$ such that (see Brockwell & Davis, 1991, p. 88)

$$\varepsilon_t = \Theta(L)^{-1} y_t = \sum_{\ell=-\infty}^{\infty} \Psi_\ell y_{t+\ell}.$$

Recoverability (cont.)

- Recoverability is a more plausible assumption than invertibility. Some DSGE models with non-invertible shocks (given standard observables) satisfy recoverability, e.g., all models with $n = n_\varepsilon$.
- DSGE models with **news shocks** assume agents receive information about changes to fundamentals in the future. If the econometrician does not observe these news directly, these models often have non-invertible representations (unless we can find very "forward-looking" data series without additional noise).
- But news shocks may be recoverable, as events *following* the arrival of the news help distinguish these shocks from other types of shocks.
- Similar discussion applies to DSGE models with **noise shocks**: the noise component of signals that agents receive about future shocks.

Proxy-SVMA: identification under recoverability

$$
\begin{aligned}
y_t &= \Theta(L)\varepsilon_t, \\
z_t &= \alpha\varepsilon_{1,t} + v_t, \\
(\varepsilon_t', v_t)' &\sim WN(0, \mathrm{diag}(1, \dots, 1, \sigma_v^2))
\end{aligned}
$$

- Assume that ε_1 is recoverable, which means that

$$\varepsilon_{1,t}^{\dagger} \equiv E^*(\varepsilon_{1,t} \mid \{y_\tau\}_{-\infty<\tau<\infty}) = \varepsilon_{1,t}.$$

- Then

$$z_t^{\dagger} \equiv E^*(z_t \mid \{y_\tau\}_{-\infty<\tau<\infty}) = \alpha\varepsilon_{1,t}^{\dagger} + 0 = \alpha\varepsilon_{1,t},$$

  since v_t is uncorr. with $\{y_t\}$ at all leads/lags. Thus,

$$\mathrm{Var}(z_t^{\dagger}) = \alpha^2 \,\mathrm{Var}(\varepsilon_{1,t}) = \alpha^2, \quad \sigma_v^2 = \mathrm{Var}(z_t) - \mathrm{Var}(z_t^{\dagger}).$$

- The shock is also identified: $\varepsilon_{1,t} = \frac{1}{\alpha} z_t^{\dagger}$.

Proxy-SVMA: identification under recoverability (cont.)

- Since α and $\varepsilon_{1,t}$ are identified, our previous results tell us that absolute IRs (and thus also historical decompositions) wrt. the first shock are also identified.
- Can we point-identify the FVD? Unfortunately, no.
  - Reason: The IV doesn't tell us how non-invertible the *other* shocks may be, which is important for the FVD.
- However, Plagborg-Moller & Wolf (2022) show that we can point-identify a modified version of the FVD that may also be of economic interest.

Testing invertibility

- Without an IV, we know that it is impossible to test for invertibility in the SVMA model. But the IV gives us a testable implication.

- Under invertibility, for all i,

$$E^*(y_{i,t} \mid \{z_\tau, y_\tau\}_{-\infty<\tau<t}) = E^*(y_{i,t} \mid \{y_\tau\}_{-\infty<\tau<t}),$$

  i.e., the IV $\{z_t\}$ *does not* Granger-cause any of the variables in y_t (conditional on all other var's).

  - Reason: $z_t = \alpha\varepsilon_{1,t} + v_t$. The second term is useless for forecasting y. Under invertibility, the information in the first term is already captured by $\{y_\tau\}_{\tau\leq t}$.

- Suggests a simple test of invertibility: test for Granger causality (e.g., using VAR methods).

- Plagborg-Moller & Wolf (2022) show that this is the *only* testable implication of invertibility. They also propose a test of recoverability.

SVAR-IV

- We have seen that we don't need to assume invertibility to identify most objects of interest using an IV, recoverability is enough (and not even needed for relative IRFs).
- But historically, many papers did assume invertibility, so we shall briefly consider this case.
- Invertibility of all shocks immediately implies the **SVAR-IV** (also called **proxy-SVAR**) model

$$A(L)y_t = H\varepsilon_t, \quad \varepsilon_t \sim WN(0, I_n),$$

$$E(\varepsilon_{1,t} z_t) \neq 0, \quad E(\varepsilon_{j,t} z_t) = 0,\ j \geq 2.$$

- Unlike earlier, we do not need to assume $E(z_t \varepsilon_\tau) = 0_{n \times 1}$ for $\tau \neq t$ in the following.

SVAR-IV: identification

$$A(L)y_t = H\varepsilon_t, \quad \varepsilon_t \sim WN(0, I_n),$$
$$E(\varepsilon_{1,t}z_t) \neq 0, \quad E(\varepsilon_{j,t}z_t) = 0,\ j \geq 2.$$

- As always, $A(L)$ is identified, as are the forecast errors

$$\eta_t \equiv A(L)y_t = H\varepsilon_t.$$

- Assume H is non-singular. In particular, $n_\varepsilon = n$.
- Then the shock ε_1 itself is identified:

$$E^*(z_t \mid \eta_t) = E^*(z_t \mid H\varepsilon_t) = E^*(z_t \mid \varepsilon_t) = E^*(z_t \mid \varepsilon_{1,t}) = \alpha\varepsilon_{1,t}.$$

- Can further identify $\alpha^2 = \mathrm{Var}(E^*(z_t \mid \eta_t))$, since $\mathrm{Var}(\varepsilon_{1,t}) = 1$.

SVAR-IV: identification (cont.)

$$\eta_t \equiv A(L)y_t = H\varepsilon_t, \quad \varepsilon_{1,t} \propto E^*(z_t \mid \eta_t)$$

- We cannot hope to identify all of H, since our IV only provides information about ε_1. But we can identify the first column $H_{\bullet 1}$.
- Exercise: Prove that

$$H_{\bullet 1} = \pm(\gamma' \Sigma^{-1} \gamma)^{-1/2} \gamma, \quad \gamma \equiv E(\eta_t z_t), \quad \Sigma \equiv \mathrm{Var}(\eta_t).$$

  (The sign is pinned down by the normalization $\alpha > 0$, say.)
- Given $H_{\bullet 1}$ and $\varepsilon_{1,t}$, we can use the usual SVAR formulas to compute:
  - *Absolute* impulse responses $\Theta_{i,1,\ell}$.
  - $FVD_{i,1,\ell}$.
  - $HD_{i,1,t}$.

SVAR-IV: estimation and inference

- SVAR-IV inference can proceed almost as easily as inference for other SVAR identification schemes.
- Given VAR residuals $\hat{\eta}_t$, can estimate $\hat{\gamma} = \frac{1}{T}\sum_{t=1}^{T} \hat{\eta}_t z_t$. $\hat{\varepsilon}_{1,t}$ given by projection of z_t onto $\hat{\eta}_t$ (normalized to variance 1).
- For delta method inference, we need to take into account the uncertainty in $\hat{\gamma}$. Can set up as GMM with the other VAR parameters.
  - For bootstrap or Bayesian approaches, it's common to assume a parametric time series model for z_t.
  - Weak-IV-robust inference: Montiel Olea, Stock & Watson (2021).
- The invertibility-based SVAR-IV estimator is sometimes called "external IV" to contrast with our earlier non-invertibility-robust "internal IV" estimator that puts z_t directly in a recursive SVAR.

Application: high-freq. identification of monetary shocks

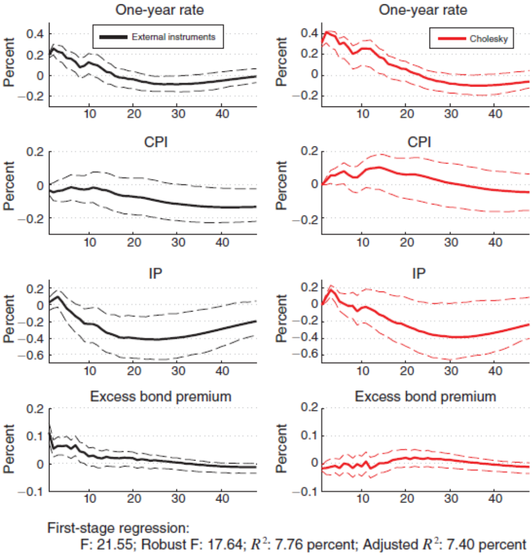

Figure 1. One-Year Rate Shock with Excess Bond Premium

Source: Gertler & Karadi (2015)

Multiple instruments

- Suppose we have more than one IV for ε_1 available:

$$z_{b,t} = \alpha_b \varepsilon_{1,t} + v_{b,t}, \quad b = 1, \ldots, r.$$

- We typically don't want to assume that $\mathrm{Cov}(v_{b,t}, v_{\tilde{b},t}) = 0$ for $b \neq \tilde{b}$, since IVs are often constructed in similar ways.

- Multiple IVs in SVAR-IV: Common to just identify

$$\varepsilon_{1,t} \propto E^*(\eta_{j,t} \mid z_{1,t}, \ldots, z_{r,t})$$

  for a single choice of j (Gertler & Karadi, 2015). Could test over-identifying restrictions by varying j and comparing.

- Multiple IVs w/o invertibility: Plagborg-Moller & Wolf (2022).

Identification via recoverability without instruments

- We have seen that recoverability is a very useful assumption if we have access to instruments/proxies.
- But the concept of recoverability can also be exploited in the absence of instruments. We then require lots of other exclusion restrictions to achieve point identification due to the severe identification issues in SVMA models.
- Examples: Mertens & Ravn (2010); Forni, Gambetti, Lippi & Sala (2017); Chahrour & Jurado (2022).

Appendix: Proxy-SVMA estimation under recoverability

- The identification formulas for Proxy-SVMA under recoverability involve various projections. How to estimate in practice?
- Simple idea:
  - Estimate a *reduced-form* VAR(p) model in the data $w_t \equiv (z_t, y_t')'$:

    $$w_t = \hat{A} w_{t-1} + \hat{\eta}_t, \quad \widehat{\text{Var}}(\hat{\eta}) = \hat{\Sigma}.$$

    (Here VAR(1) for simplicity.) Note: We're not assuming an SVAR, just approximating the ACF of the data.
  - Compute objects like $z_t^{\dagger} \approx E^*(z_t \mid \{y_\tau\}_{t-K<\tau<t+K})$ and $\text{Var}(z_t^{\dagger}) \approx \text{Var}(z_t) - \text{Var}^*(z_t \mid \{y_\tau\}_{t-K<\tau<t+K})$ using the Kalman smoother applied to the artificial state space model

    $$\begin{aligned} y_t &= (0_{n\times 1}, I_n) w_t, \\ w_t &= \hat{A} w_{t-1} + \xi_t, \\ \xi_t &\sim N(0_{(n+1)\times 1}, \hat{\Sigma}). \end{aligned}$$

Appendix: Proxy-SVMA estimation under recov'y (cont.)

- Don't get confused by the state space model: z_t is in fact observed data, but since we are interested in expressions like $E^*(z_t \mid \{y_\tau\}_{t-K<\tau<t+K})$, we use a Kalman smoother that treats z_t as a latent state (component of w_t).
- Inference: bootstrap the reduced-form VAR, or do Bayesian inference on the reduced-form VAR.

Appendix: Partial identification in Proxy-SVMA model

- Recall the Proxy-SVMA model:

$$
\begin{aligned}
y_t &= \Theta(L)\varepsilon_t, \\
z_t &= \alpha\varepsilon_{1,t} + v_t, \\
(\varepsilon_t', v_t)' &\sim WN(0, \operatorname{diag}(1, \dots, 1, \sigma_v^2)).
\end{aligned}
$$

- Earlier slides:
  - If we knew α, we would be able to identify the absolute impulse responses $\Theta_{i,1,\ell} = \frac{1}{\alpha}\operatorname{Cov}(y_{i,t}, z_{t-\ell})$ wrt. the first shock.
  - If we assume recoverability, then we can point-identify α.
- If we do not assume recoverability *a priori*, then α is only partially identified: We don't know the signal-to-noise ratio α^2/σ_v^2 of the IV.

Appendix: Partial identif'n in Proxy-SVMA model (cont.)

$$y_t = \Theta(L)\varepsilon_t, \quad z_t = \alpha\varepsilon_{1,t} + v_t$$

- Upper bound for α^2:

$$\alpha^2 \le \alpha^2 + \sigma_v^2 = \mathrm{Var}(z_t).$$

- Lower bound for α^2:
  - Recall notation $z_t^\dagger \equiv E^*(z_t \mid \{y_\tau\}_{-\infty<\tau<\infty})$, and similarly for $\varepsilon_{1,t}^\dagger$.
  - It can be shown that (Brockwell & Davis, 1991, p. 439)

    $$s_{\varepsilon_1^\dagger}(\omega) \le s_{\varepsilon_1}(\omega) \quad \text{for all } \omega.$$

    Intuitively, "explained sum of squares" $\le$ "total sum of squares".
  - Hence, since $z_t^\dagger = \alpha\varepsilon_{1,t}^\dagger$,

$$\forall\, \omega \in [0,\pi]\colon \;\; s_{z^\dagger}(\omega) = \alpha^2 s_{\varepsilon_1^\dagger}(\omega) \le \alpha^2 s_{\varepsilon_1}(\omega) = \alpha^2 \times \tfrac{1}{2\pi},$$

$$\Longrightarrow \alpha^2 \ge 2\pi \sup_{\omega\in[0,\pi]} s_{z^\dagger}(\omega).$$

Appendix: Partial identif'n in Proxy-SVMA model (cont.)

- Plagborg-Moller & Wolf (2022) show that these bounds are *sharp*. That is, the *identified set* for α^2 is the interval

$$\left[2\pi \sup_{\omega \in [0,\pi]} s_{z^\dagger}(\omega), \mathrm{Var}(z_t)\right].$$

- Using the identif. set for α, we can now compute identif. sets for impulse responses. These are also interval-identified.